

G.SURENDHRA (1925254310)

1. Create Database

CREATE DATABASE EmployeeDB;

USE EmployeeDB;

```
mysql> INSERT INTO Employee
-> (Eid, Ename, DeptId, Designation, Salary, DOJ)
-> VALUES
-> (101, 'Sudha', 'D2', 'Clerk', 20000, '2010-04-01'),
-> (102, 'David', 'D1', 'Manager', 50000, '2018-02-18'),
-> (103, 'Kiran', 'D1', 'Salesman', 20000, '2014-03-07');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Employee;
+-----+-----+-----+-----+-----+-----+
| Eid | Ename | DeptId | Designation | Salary | DOJ |
+-----+-----+-----+-----+-----+-----+
| 101 | Sudha | D2     | Clerk       | 20000.00 | 2010-04-01 |
| 102 | David | D1     | Manager     | 50000.00 | 2018-02-18 |
| 103 | Kiran | D1     | Salesman    | 20000.00 | 2014-03-07 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM Employee
-> WHERE Salary > (
-> SELECT AVG(Salary)
-> FROM Employee
-> );
+-----+-----+-----+-----+-----+-----+
| Eid | Ename | DeptId | Designation | Salary | DOJ |
+-----+-----+-----+-----+-----+-----+
| 102 | David | D1     | Manager     | 50000.00 | 2018-02-18 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

2. Create Department Table

Table: Department

DeptId Dname

D1 Sales

D2 Marketing

D3 Finance

SQL Query

```
CREATE TABLE Department (
  DeptId VARCHAR(10) PRIMARY KEY,
  Dname VARCHAR(50) NOT NULL
);
```

Constraints

- DeptId → Primary Key
- Dname → NOT NULL

```
mysql> SELECT *
-> FROM Employee
-> WHERE Salary > (
-> SELECT AVG(Salary)
-> FROM Employee
-> );
```

Eid	Ename	DeptId	Designation	Salary	DOJ
102	David	D1	Manager	50000.00	2018-02-18

1 row in set (0.00 sec)

```
mysql> SELECT E.Eid,
-> E.Ename,
-> D.Dname
-> FROM Employee E
-> JOIN Department D
-> ON E.DeptId = D.DeptId;
```

Eid	Ename	Dname
101	Sudha	Marketing
102	David	Sales
103	Kiran	Sales

3 rows in set (0.01 sec)

3. Insert Department Data

```
INSERT INTO Department (DeptId, Dname)
VALUES
('D1', 'Sales'),
('D2', 'Marketing'),
('D3', 'Finance');
```

```
mysql> SELECT *
-> FROM Employee
-> ORDER BY Salary DESC;
```

Eid	Ename	DeptId	Designation	Salary	DOJ
102	David	D1	Manager	50000.00	2018-02-18
101	Sudha	D2	Clerk	20000.00	2010-04-01
103	Kiran	D1	Salesman	20000.00	2014-03-07

3 rows in set (0.00 sec)

4. Create Employee Table

Table: Employee

Eid	Ename	DeptId	Designation	Salary	DOJ
101	Sudha	D2	Clerk	20000	01-Apr-2010
102	David	D1	Manager	50000	18-Feb-2018
103	Preethi	D3	Clerk	35000	13-Jun-2011
104	Kiran	D1	Salesman	20000	07-Mar-2014

Eid Ename DeptId Designation Salary DOJ

105	Meenal	D2	Clerk	50000	09-Dec-2011
106	Sunitha	D3	Manager	60000	25-Sep-2018
107	Akhil	D3	Clerk	25000	14-Feb-2016
108	Sushma	D2	Manager	45000	31-Jan-2012

SQL Query

```
CREATE TABLE Employee (  
  Eid INT PRIMARY KEY,  
  Ename VARCHAR(50),  
  DeptId VARCHAR(10),  
  Designation VARCHAR(50),  
  Salary DECIMAL(10,2),  
  DOJ DATE,  
  FOREIGN KEY (DeptId) REFERENCES Department(DeptId),  
  CHECK (Salary >= 10000)  
);
```

Constraints

- Eid → **Primary Key**
- DeptId → **Foreign Key**
- Salary → **Not less than 10000**
- Department.DeptId → Referenced by Employee.DeptId

```
mysql> SELECT DISTINCT Designation  
-> FROM Employee;  
  
+-----+  
| Designation |  
+-----+  
| Clerk       |  
| Manager     |  
| Salesman    |  
+-----+  
  
3 rows in set (0.00 sec)
```

5. Insert Employee Data

```
INSERT INTO Employee  
(Eid, Ename, DeptId, Designation, Salary, DOJ)  
VALUES  
(101, 'Sudha', 'D2', 'Clerk', 20000, '2010-04-01'),  
(102, 'David', 'D1', 'Manager', 50000, '2018-02-18'),  
(103, 'Preethi', 'D3', 'Clerk', 35000, '2011-06-13'),  
(104, 'Kiran', 'D1', 'Salesman', 20000, '2014-03-07'),  
(105, 'Meenal', 'D2', 'Clerk', 50000, '2011-12-09'),  
(106, 'Sunitha', 'D3', 'Manager', 60000, '2018-09-25'),  
(107, 'Akhil', 'D3', 'Clerk', 25000, '2016-02-14'),  
(108, 'Sushma', 'D2', 'Manager', 45000, '2012-01-31');
```

```
mysql> SELECT *
-> FROM Employee
-> ORDER BY DeptId ASC,
-> Salary ASC;
```

Eid	Ename	DeptId	Designation	Salary	DOJ
103	Kiran	D1	Salesman	20000.00	2014-03-07
102	David	D1	Manager	50000.00	2018-02-18
101	Sudha	D2	Clerk	20000.00	2010-04-01

3 rows in set (0.00 sec)

QUESTION 1

Question

Write SQL query to display all the employees who earn more than average salary of all the employees in the company.

Concept

Aggregate Function + Subquery

We use AVG() to calculate the average salary.

Query

```
SELECT *
FROM Employee
WHERE Salary > (
    SELECT AVG(Salary)
    FROM Employee
);
```

Explanation

- AVG(Salary) → Calculates average salary.
- The outer query selects employees whose salary is greater than that average.

Expected Result

Average salary = **38,125**

Employees earning more than average:

Eid Ename Salary

102 David 50000

105 Meenal 50000

106 Sunitha 60000

108 Sushma 45000

QUESTION 2

Question

Write SQL query to display the fields Eid, Ename and Dname.

Concept

JOIN

Eid and Ename are in Employee, while Dname is in Department.

Query

```
SELECT E.Eid,
       E.Ename,
       D.Dname
FROM Employee E
JOIN Department D
ON E.DeptId = D.DeptId;
```

Explanation

- E → Alias for Employee.
- D → Alias for Department.
- JOIN → Combines the two tables.
- ON → Matches the DeptId values.

Expected Result

Eid Ename Dname

101 Sudha Marketing
 102 David Sales
 103 Preethi Finance
 104 Kiran Sales
 105 Meenal Marketing
 106 Sunitha Finance
 107 Akhil Finance
 108 Sushma Marketing

QUESTION 3

Question

Write SQL query to sort the Employee table in descending order of salaries.

Concept

ORDER BY

Query

```
SELECT *
FROM Employee
ORDER BY Salary DESC;
```

Explanation

- ORDER BY → Sorts the records.
- DESC → Descending order.
- Highest salary appears first.

Expected Salary Order

Ename Salary

Sunitha 60000
 David 50000
 Meenal 50000
 Sushma 45000
 Preethi 35000
 Akhil 25000
 Sudha 20000
 Kiran 20000

QUESTION 4

Question

Write SQL query to list all the job designations in the Employee table without repetitions.

Concept

DISTINCT

Query

```
SELECT DISTINCT Designation
FROM Employee;
```

Explanation

DISTINCT removes duplicate values.

Expected Result

Designation

Clerk

Designation

Manager

Salesman

QUESTION 5

Question

Write SQL query to display all the employee details Department-wise and in the ascending order of their salaries.

Concept

ORDER BY Multiple Columns

Query

```
SELECT *  
FROM Employee  
ORDER BY DeptId ASC,  
Salary ASC;
```

Explanation

- First, employees are arranged according to DeptId.
- Then salaries are arranged from lowest to highest within each department.
- ASC means ascending order.

Department order

D1 → Sales

D2 → Marketing

D3 → Finance

QUESTION 6

Question

Write SQL query to display all the clerks in DeptId D2.

Concept

WHERE + AND

Query

```
SELECT *  
FROM Employee  
WHERE Designation = 'Clerk'  
AND DeptId = 'D2';
```

Explanation

Two conditions must be satisfied:

1. Designation must be Clerk.
2. Department must be D2.

Expected Result

Eid	Ename	DeptId	Designation	Salary
101	Sudha	D2	Clerk	20000

```
mysql> SELECT *  
-> FROM Employee  
-> WHERE Designation = 'Clerk'  
-> AND DeptId = 'D2';  
+----+-----+-----+-----+-----+-----+  
| Eid | Ename | DeptId | Designation | Salary | DOJ |  
+----+-----+-----+-----+-----+-----+  
| 101 | Sudha | D2     | Clerk       | 20000.00 | 2010-01-01 |  
+----+-----+-----+-----+-----+-----+  
1 row in set (0.00 sec)
```

Eid	Ena me al	DeptI d	Designati on	Salar y
				0

QUESTION 7

Question

Write SQL query to display all the employees who joined in the year 2011.

Concept

YEAR() Function

Query

```
SELECT *
FROM Employee
WHERE YEAR(DOJ) = 2011;
```

Explanation

YEAR(DOJ) extracts the year from the Date of Joining.

Expected Result

Eid Ename DOJ

```
103 Preethi 2011-06-13
105 Meenal 2011-12-09
```

QUESTION 8

Question

Write SQL query to display all the employees who joined in the month of February.

Concept

MONTH() Function

Query

```
SELECT *
FROM Employee
WHERE MONTH(DOJ) = 2;
```

Explanation

- MONTH(DOJ) extracts the month number.
- February = 2.

Expected Result

Eid Ename DOJ

```
102 David 2018-02-18
107 Akhil 2016-02-14
```

QUESTION 9

Question

Write SQL query to display all the employees whose salary is between 30000 and 45000.

Concept

BETWEEN Operator

Query

```
SELECT *
FROM Employee
WHERE Salary BETWEEN 30000 AND 45000;
```

Explanation

BETWEEN selects values within a specified range.

Here:

Minimum = 30000

Maximum = 45000

Both 30000 and 45000 are included.

Expected Result

Eid Ename Salary

Eid Ename Salary

103 Preethi 35000

108 Sushma 45000

Note: Akhil's salary is 25,000, so he is not included.

```
mysql> SELECT *  
-> FROM Employee  
-> WHERE MONTH(DOJ) = 2;
```

Eid	Ename	DeptId	Designation	Salary	DOJ
102	David	D1	Manager	50000.00	2018-02-18

1 row in set (0.00 sec)

QUESTION 10

Question

Write SQL query to display all the employee details along with their work experience in the company till current date.

Concept

Date Functions

Query

```
SELECT Eid,  
       Ename,  
       DeptId,  
       Designation,  
       Salary,  
       DOJ,  
       TIMESTAMPDIFF(YEAR, DOJ, CURDATE()) AS Work_Experience  
FROM Employee;
```

Explanation

- DOJ → Date of joining.
- CURDATE() → Current date.
- TIMESTAMPDIFF() → Calculates the difference between two dates.
- YEAR → Gives completed years.
- AS Work_Experience → Gives a name to the calculated column.

Expected Output Format

Eid	Ename	DOJ	Work_Experience
-----	-------	-----	-----------------

101	Sudha	2010-04-01	calculated
-----	-------	------------	------------

102	David	2018-02-18	calculated
-----	-------	------------	------------

103	Preethi	2011-06-13	calculated
-----	---------	------------	------------

104	Kiran	2014-03-07	calculated
-----	-------	------------	------------

105	Meenal	2011-12-09	calculated
-----	--------	------------	------------

106	Sunitha	2018-09-25	calculated
-----	---------	------------	------------

107	Akhil	2016-02-14	calculated
-----	-------	------------	------------

108	Sushma	2012-01-31	calculated
-----	--------	------------	------------

The exact experience changes automatically because CURDATE() uses the **current date**.

★ Important SQL Commands Used

Q.No	SQL Concept	Main Command
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1	Average + Subquery	AVG()
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2	Combining tables	JOIN
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Q.No	SQL Concept	Main Command
3	Descending sorting	ORDER BY DESC
4	Remove duplicates	DISTINCT
5	Multiple sorting	ORDER BY ASC
6	Multiple conditions	WHERE, AND
7	Extract year	YEAR()
8	Extract month	MONTH()
9	Range checking	BETWEEN
10	Calculate experience	TIMESTAMPDIFF()

```
mysql> SELECT *
-> FROM Employee
-> WHERE Salary BETWEEN 30000 AND 45000;
Empty set (0.00 sec)
```

```
mysql> SELECT Eid,
-> Ename,
-> DeptId,
-> Designation,
-> Salary,
-> DOJ,
-> TIMESTAMPDIFF(YEAR, DOJ, CURDATE()) AS Work_Experience
-> FROM Employee;
```

Eid	Ename	DeptId	Designation	Salary	DOJ	Work_Experience
101	Sudha	D2	Clerk	20000.00	2010-04-01	16
102	David	D1	Manager	50000.00	2018-02-18	8
103	Kiran	D1	Salesman	20000.00	2014-03-07	12

```
3 rows in set (0.05 sec)
```